

Evidence for the Jelly Ocean Hypothesis in the Northwest Atlantic

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Abstract

The Jelly Ocean Hypothesis posits that gelatinous zooplankton populations are increasing worldwide as a result of changing ocean conditions. This hypothesis has been the subject of debate in ocean science. We examined the Northeast US Continental Shelf for increases in the occurrence of gelatinous zooplankton taxa using a 50-year survey and a species distribution model. Time series analysis showed statistically significant increases in salps, siphonophores, and coelenterates (scyphozoans) in late summer and early autumn across the entire study region. Species distribution models showed that the strongest covariates associating with occurrence of gelatinous taxa were day length, depth, and surface temperature for coelenterates; day length, surface and bottom temperature, and oxygen for salps; and day length, depth, bottom temperature, and oxygen for siphonophores. The results form one piece of evidence in favor of a long-term rise in gelatinous species. The roughly synchronized increases in these widely diverse taxa suggest that changes in environmental or ecological conditions favor a gelatinous body plan generally.

Introduction

In the study of zooplankton ecology, there has often been a “crustaceocentric” bias (Haddock 2004). The study of gelatinous species, collectively termed gelata, has been underemphasized as compared to crustaceans. In the early 2000s, gelata moved into the spotlight, following conspicuous blooms that damaged fisheries (Kideys et al. 2005), aquaculture (Clinton et al. 2021), energy production (Wang et al. 2023), and recreation (Nunes et al. 2015). In addition to motivating new research on gelata, these events raised the question of whether there was a larger pattern of change.

Emerging from a series of studies was the Jelly Ocean Hypothesis, which posits that gelatinous zooplankton populations are increasing worldwide as a result of changing ocean conditions, primarily anthropogenic. These studies argued that due a multiplicity of causes, abundance of gelatinous species is growing. Proposed causes are primarily human-driven, including warming (ref), eutrophication, translocation (Kideys et al. 2005), overfishing, bottom trawling (Qui 2014), pollution, aquaculture, and acidification (Richardson and Gibbons 2008). There is also evidence to suggest that shifts toward dominance of gelata are difficult to reverse because of dynamics like hysteresis (Richardson et al. 2009). The Jelly Ocean Hypothesis echoes a concern that has been around for centuries; the 19th century novelist Jules Verne once predicted that, “crowded with jellyfish, squid, and other devilfish, the oceans will have become huge centers of infection” (REF). While the term “infection” might not be fair to jellyfish, who simply inhabit the ocean, the potential influence on human industry and recreation make the question worthy of examination.

Testing the Jelly Ocean Hypothesis requires long time series (>30 years) of gelata abundances in order to compare to historical levels. The sporadic nature of such datasets has left the Jelly Ocean Hypothesis largely unsupported (Condon et al. 2012). In lieu of time series, studies have used qualitative data such as news reports to argue that gelata are increasing globally (Brotz et al. 2012). However, it is difficult to distinguish these perceived increases from multi-annual or multi-decadal fluctuations. A recent meta-analysis of time series showed that most perceived increases appear to reflect a ~20 year cyclicity, with a worldwide upswing in the 1990s that coincided with the increase in concern over gelata (Condon et al. 2013), although some

notable long-term increases in gelata were not included in the analysis (Atkinson et al. 2004). A metaanalysis of a corpus of research on this topic found flawed citations and misinterpreted conclusions (Sanz-Martin et al. 2016), and at this point, the Jelly Ocean Hypothesis remains very much an open question.

In some ways, the Jelly Ocean Hypothesis is loosely posed. Gelata are taxonomically very diverse, encompassing multiple phyla. This diversity confounds attempts to quantify increasing trends. The list of species classified under the gelatinous umbrella varies based on the source, and includes taxa with drastic and important differences in their life histories, trophic positions, and ecological roles. For example, pelagic ctenophores are primarily copepod predators, whereas pelagic tunicates graze some of the smallest microbes. Cnidarians and ctenophores are the two most common taxa to group together in analysis, and yet they are phylogenetically very distant (Moroz et al. 2014). Perhaps the only characteristic common to the many gelatinous phyla is a body with a high water content, alternatively expressed as a low carbon:volume ratio. There is no *a priori* reason to expect that phyla with such striking differences and perhaps only a superficial similarity should be grouped together with respect to their temporal trends. Some have suggested that the Jelly Ocean Hypothesis is ill-posed—i.e. which species are increasing, where, and for what reason (Gibbons & Richardson 2013)?

On the other hand, despite their major differences, there is some evidence that a diverse collection of gelata can vary together (Condon et al. 2013). This suggests that certain conditions can favor the gelatinous body form in general over other strategies that utilize a high carbon:volume ratio (e.g. crustaceans). This reframes the Jelly Ocean Hypothesis in trait-based terms: conditions in the ocean ecosystem can shift to benefit the gelatinous body form. This would be inclusive of gelata in general, despite differences in phylogeny, trophic position, or other life history traits.

We revisited the Jelly Ocean Hypothesis using a 50-year spatial survey on the northwest North Atlantic Shelf. We addressed two main questions: 1) what are the temporal changes in gelata presence; and 2) what are the underlying conditions associated with changes. We examined three taxonomic groups well-represented in the dataset—siphonophores, salps, and coelenterates—to explore the idea that certain conditions can favor the gelatinous body plan in general.

Methods

Survey Data

The Northeast Fisheries Science Center Ecosystem Monitoring (ECOMON) program has collected seasonal zooplankton measurements since 1977 as part of its mission to monitor and assess the Northeast Continental Shelf ecosystem (Figure 1). Sampling includes oblique tows using a 61 cm bongo net fitted with a 333 μm mesh at stratified locations across the shelf (Figure 1). Tows are a minimum of five minutes in duration and sample from the surface to within 5 m of the seabed or to a maximum depth of 200 m. A mechanical flowmeter fitted in the mouth of each net is used to quantify volume sampled. Gelatinous zooplankton are identified into broad taxonomic groups, including siphonophores, ctenophores, salps, and coelenterates (i.e. scyphozoans). The broad grouping, while missing taxonomic detail, has allowed for internal consistency over the duration of the survey. Because of the absence of high taxonomic resolution, or any information on size of animals in the samples, The analysis conducted here uses presence or absence of each taxonomic group.

Time series analysis

The ECOMON survey has varied in frequency over its duration. To correct for seasonal aliasing and spatial bias, we followed the common approach of grouping by large spatial polygons that represent different habitats, and analyzed the time series by month. Polygons include: Gulf of Maine, Georges Bank, New York Bight, and Mid Atlantic Bight (Walsh et al. 2015). Because of the low taxonomic resolution of gelatinous taxa and the bulk biomass quantification, we viewed measurements in a presence-absence context, tracking the proportion of samples in a region and in a time period that contain presences. For each month and within

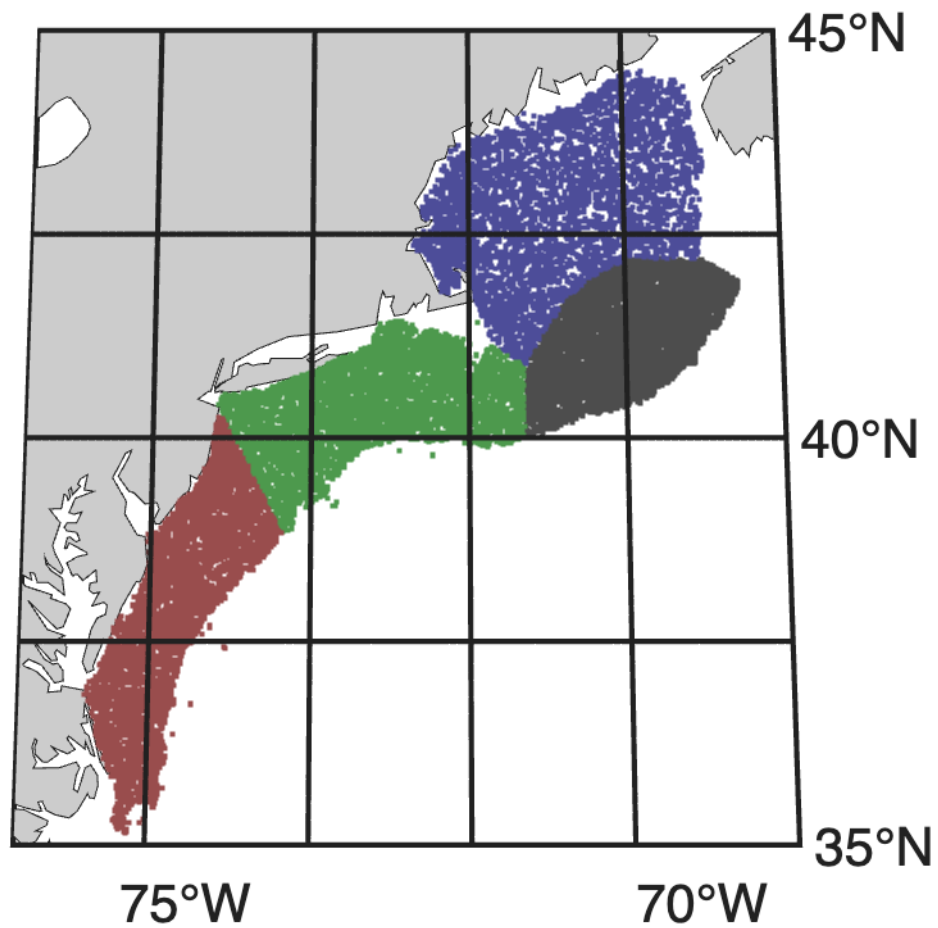


Figure 1: Map of the study region. Points indicate sampling locations colored by the four polygons: Gulf of Maine (blue), Georges Bank (grey), Southern New England (green), and Mid-Atlantic Bight (red).

each region, we tested for statistically significant increases over time using a bivariate correlation test against year.

Spatial analysis

To test for environmental associations with high or low occurrence of gelatinous taxa across the study domain, we developed a species distribution model for each taxonomic group. We used physical covariates from the Global Ocean Physics Reanalysis (GLORYS) and biogeochemical covariates the Operational Mercator Ocean biogeochemical global ocean analysis (NEMO) obtained from the Copernicus repository (copernicus.eu), which provide daily temporal resolution and 1/12 and 1/4 spatial degree resolution, respectively. Physical covariates included: sea surface temperature, sea bottom temperature, sea surface salinity, mixed layer thickness, and surface velocity magnitude. Biogeochemical covariates included: surface chlorophyll, surface net primary productivity, surface oxygen, surface nitrate, surface phosphate, and surface silicate. We also included day length and the time derivative of day length to represent seasonal dependencies. This choice was based on series of experiments testing different approaches to representing season, including using month as a factor, using a rolling seasonality index, and using the height and angle of the sun. Models using day length and the time derivative of day length had the strongest performance. We also included bathymetric depth and bathymetric slope. A common threshold for covariate removal due to cross correlation is an r-squared value of 0.7. Because removal of covariates leads to information loss and predictive power in tree-based algorithms (Hanberry 2024), we kept all covariates. However, we included this information in our interpretation and made additional model runs to test the effect of cross correlation. Only a few pairs of covariates met this threshold: bathymetric depth versus bathymetric slope (0.71), surface temperature versus surface oxygen (0.89), surface temperature versus day-length time derivative (0.79), surface chlorophyll and net primary productivity (0.81), and surface oxygen versus day-length time derivative (0.77). Because of the high correlation between sea surface temperature and surface oxygen, and the physical relationship between the two (water temperature is related to oxygen content), we ran a second set of runs using subsurface oxygen. We used a depth of 100m for subsurface oxygen, which limited the extent of the model domain.

We used a boosted regression tree algorithm to fit the full set of observations to environmental covariates. Boosted regression trees are well suited to variables that have sharp jumps or thresholds, as is the case for the presence of gelatinous zooplankton, and are less prone to overfitting in species distribution models (Elith et al. 2005). We modeled each taxon using tidymodels workflows within R, optimizing the inbuilt area-under-curve (AUC) metric (Fielding and Bell 1997, Kuhn and Wickham, 2020). We used a set of hyperparameters optimized in a previous study that used the same dataset and methodology (Johnson et al. 2025); this choice was made following a separate tuning simulation that proved the resulting models highly insensitive to hyperparameters. We used a testing split of 25%, and ran a 25-fold cross-validated model to quantify algorithmic uncertainty around the median output. Cross correlation between environmental covariates was used in the interpretation of output, but not in covariate selection because the goal of the model was to explore covariate associations. Custom-built response curves (Johnson et al. 2025) were used to interpret associations between model response and variations in environmental covariates (Elith et al., 2005).

Results

Time series analysis

Over almost five decades of measurements, there were consistent patterns of statistically significant increases in the recorded presence of gelatinous taxa (Figure 2, S1). While the survey does not cover every month in every year, there is enough coverage across seasons to see consistent patterns. Out of 144 month-region combinations, 56 (39%) indicate a statistically significant increase ($p < 0.05$) and zero (0%) indicate a statistically significant decrease. In the months August-November, 40 of the 48 time series (83%) show a statistically significant increase (Figure 2). In other words, in late summer and early fall, all taxa show an increase across the entire study region.

The results during other times of year were more mixed. On Georges Bank, there were statistically significant increases across the spring months for siphonophores and coelenterates. For the other taxa, regions, and months, there were some scattered increases but no strong pattern (Figure S1).

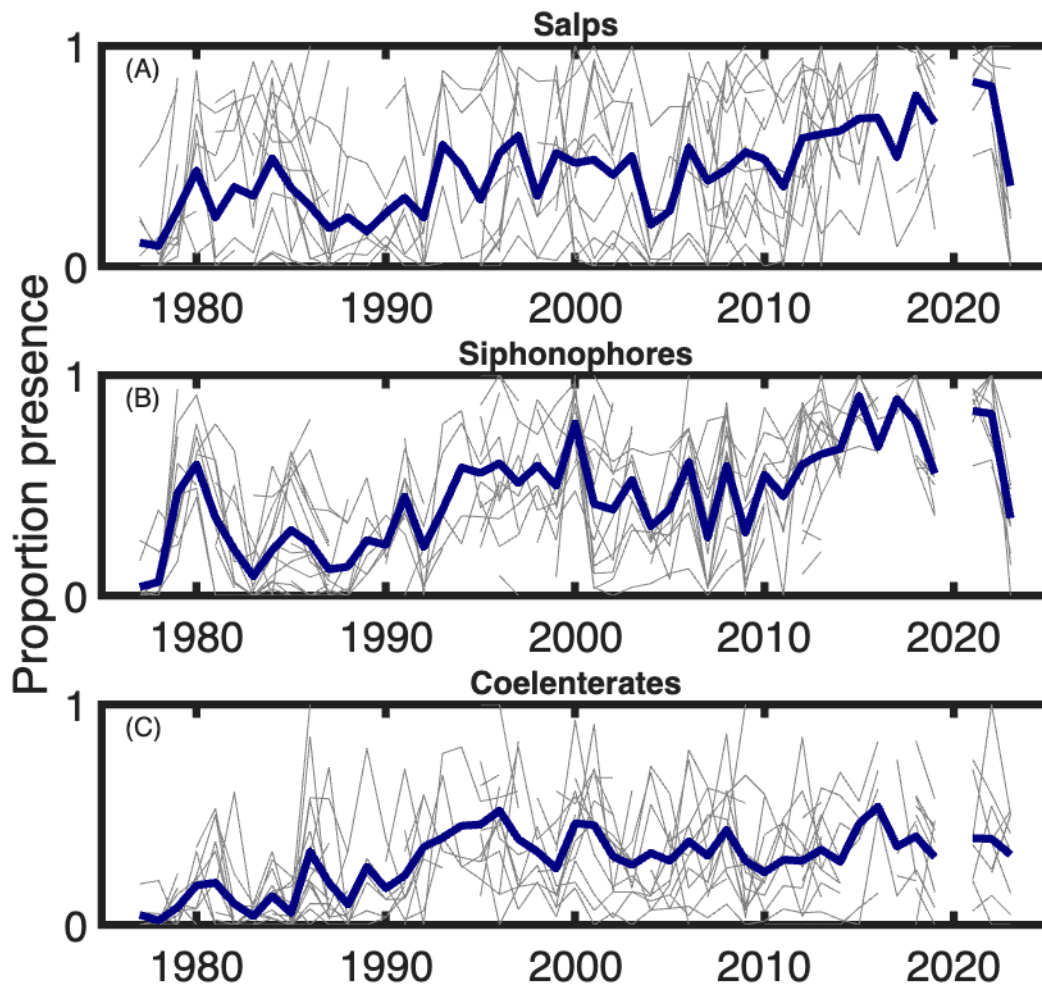


Figure 2: Time series of the proportion of presences of the indicated taxon for the months August - November (grey lines) for all regions, and the average of all shown time series (black lines). Full time series breakdowns are provided in supplementary figure S1

Spatial analysis

For the boosted regression tree distribution model, performance was generally good across the three taxonomic groups. For coelenterates, the model had an AUC value of 0.79, an accuracy of 0.77, and an F-score of 0.49. For salps, the model had an AUC value of 0.93, an accuracy of 0.89, and an F-score of 0.72. For siphonophores, the model had an AUC value of 0.83, an accuracy of 0.77, and an F-score of 0.63. The model tended to favor specificity over sensitivity over all runs.

The strongest covariates associating with occurrence of gelatinous taxa were day length, surface and bottom temperature, and oxygen for salps; day length, depth, and surface temperature for coelenterates; and day

length, depth, bottom temperature, and oxygen for siphonophores (Figure S2-4). Salps had a positive monotonic response to temperature at the surface and at depth, while the other taxa had broad modes of positive temperature response. Net primary production and mixed layer thickness were consistently the weakest covariates. For salps and siphonophores, and to some degree for coelenterates, the response ticked up for very low oxygen values. Because of the strong connection between surface oxygen and surface temperature, we ran a series of tests using oxygen at 100 m, below the mixed layer. These patterns persisted.

The three taxonomic groups had very distinct seasonal cycles, represented in the model output (Figure 3, S5-7). Salps were absent in the winter and spring, with a strong bloom in the autumn. The bloom was strongest in the Mid-Atlantic Bight, where nearly all samples contain salps, and extended northeast over Georges Bank, while presence remained generally low in the Gulf of Maine. Coelenterates were present year-round in most nearshore areas and generally absent over deeper waters, with a dip in the winter. Siphonophores had a more northerly range, with higher presence over deeper waters and a notably high presence over the continental slope in autumn.

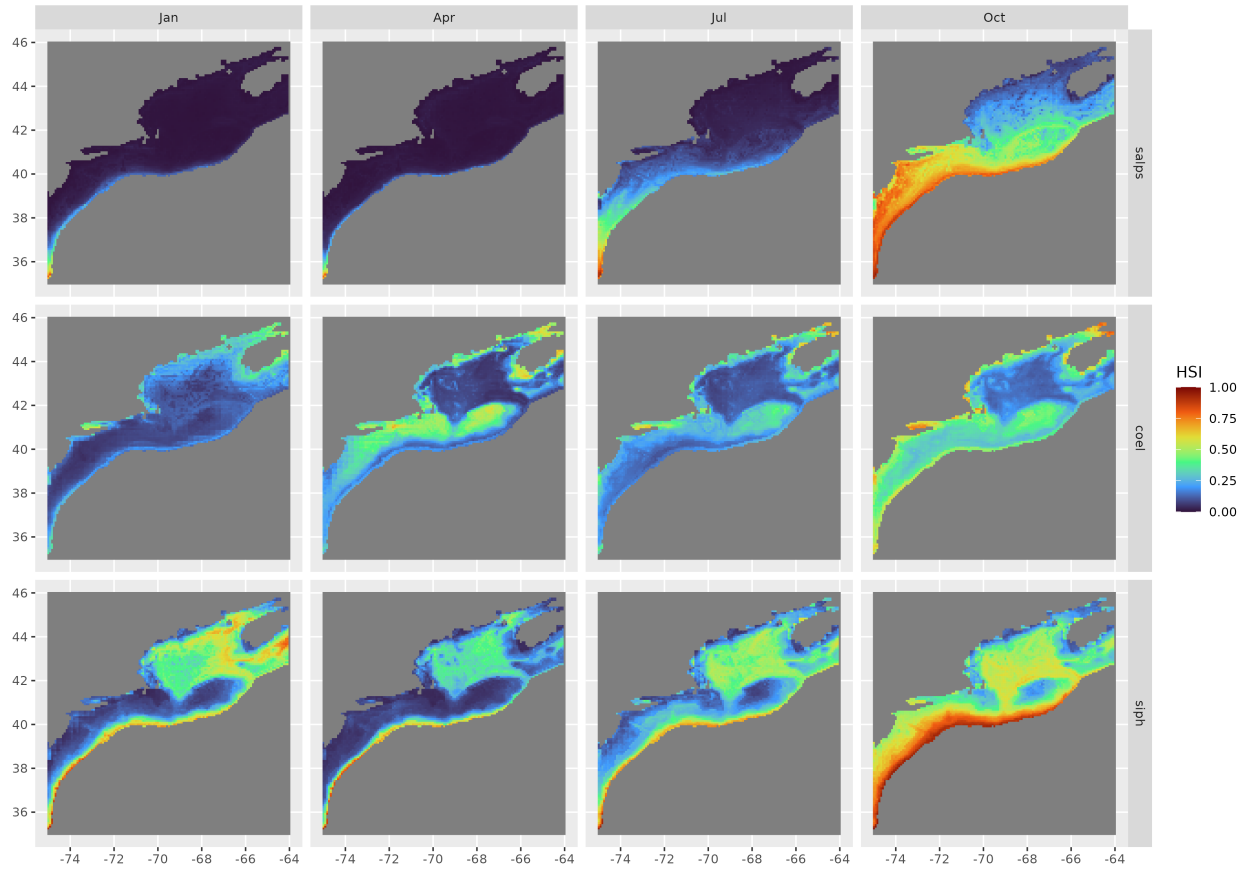


Figure 3: Model output averaged into climatologies for the three taxa, showing January, April, July, and October (columns) for salps, coelenterates, and siphonophores (rows). The full climatology for each taxon is in the supplement (S5-S7).

Discussion

Large aggregations of gelata can appear suddenly and often inspire widespread public interest (Record, Tupper, and Pershing 2018). This has happened in the northeast US, prompting the same questions that underly the Jelly Ocean Hypothesis: Are jellyfish (gelata) increasing, and if so, what is the cause? For the region considered in this study, fifty years of measurements provides clear evidence that there is a widespread increase of gelata, focused particularly in the autumn months. In some places, and for some taxa, there has been a trend from gelata being rare in samples in the early years of the survey to occurring in nearly all of the samples in the later years of the survey. This pattern is particularly pronounced for salps and siphonophores in the northern regions and the autumn months.

The question of cause is more complicated. The northeast US shelf has undergone notable oceanographic changes over the past fifty years, some of which are apparent in the computed gelata time series. There was a decadal shift among copepod species between the 1980s and the 1990s, driven by stratification (Pershing and Kemberling 2024), which appears in many of the gelata time series (Figure 2). In the Gulf of Maine in particular, a 1990s shift that ends in the early 2000s appears across taxa and is particularly pronounced for siphonophores (Figure S1). The region also underwent a decade of extremely rapid warming beginning around 2005, depending on delineation (Pershing et al. 2015). This period does not stand out in particular in the gelata time series, except again among salps in the Gulf of Maine and perhaps Georges Bank (Figure S1). Finally, the region has underwent an abrupt shift toward cold, fresh water beginning in autumn 2023 (Record, Pershing, and Rasher 2024). This is possibly reflected in the sharp drop in salp and siphonophore presence in fall of 2023, which don't appear to be driven by a sampling bias. Subsequent years of data will help clarify this possibility. Taken in total, the evidence suggests a relationship to the changing water masses of the region, but without finer taxonomic resolution, or size or life stage information, it is difficult to make more direct links.

The species distribution model can be interpreted as providing some evidence for associations with changing environmental variables. Salps showed a clear positive response to temperature, which is consistent with the expectation for the dominant species of salps known to inhabit the region – e.g. *Salpa* spp. and *Thalia* spp. (Vargas and Madin 2004; Madin et al. 2006). In the Southern Ocean, a long-term increase in salps has been associated with warmer waters and lower productivity (Atkinson et al. 2004). Neither productivity nor chlorophyll stood out as a strong covariate in this modeling exercise, but the positive association with warmer temperatures was notable, as the pattern would have to be consistent across multiple species to emerge.

Coelenterates did not have a monotonic relationship with temperature. This is not surprising, as some of the most common scyphozoan jellyfish in the range have a subarctic distribution (e.g. *Cyanea* spp.). The strongest covariates for this taxonomic group had to do with seasonality and bathymetry—i.e. variables without long-term trends. This group also had a less pronounced long-term increase, though still significant. A better understanding of the drivers of this trend would probably require more taxonomically detailed dataset. Siphonophores showed mostly positive relationships with temperature, both at the surface and at the bottom.

Salps and siphonophores, and to some degree coelenterates, had a positive response to very low oxygen values. A test using subsurface oxygen, to account for the relationship between oxygen and temperature, maintained this pattern. Because of their low carbon-to-volume ratio, gelatinous species are often able to survive at oxygen levels that are too low for their competitors or that have a slowing effect on their prey (Purcell et al. 2001). The results here are consistent with that property. One hypothesis for a cross-phylum increase in diverse gelata is that lower oxygen favors a gelatinous body plan.

Conclusion

A 50-year survey on the northeast US continental shelf showed strong evidence of an increase across diverse gelatinous zooplankton taxa. In the context of the still-debated Jelly Ocean Hypothesis, the results contribute one piece of evidence in favor of a long-term rise in gelatinous species. The roughly synchronized increases in

these widely diverse taxa suggest that changes in environmental or ecological conditions favor a gelatinous body plan generally. While there are a multiplicity of drivers of these changes, one hypothesis worth pursuing is the possibility that low oxygen favors gelatinous species generally. The survey only provided broad taxonomic groupings, which limited the analysis to some degree, but the fact that the long-term increases were apparent despite the coarse taxonomic resolution is compelling.

Supplement

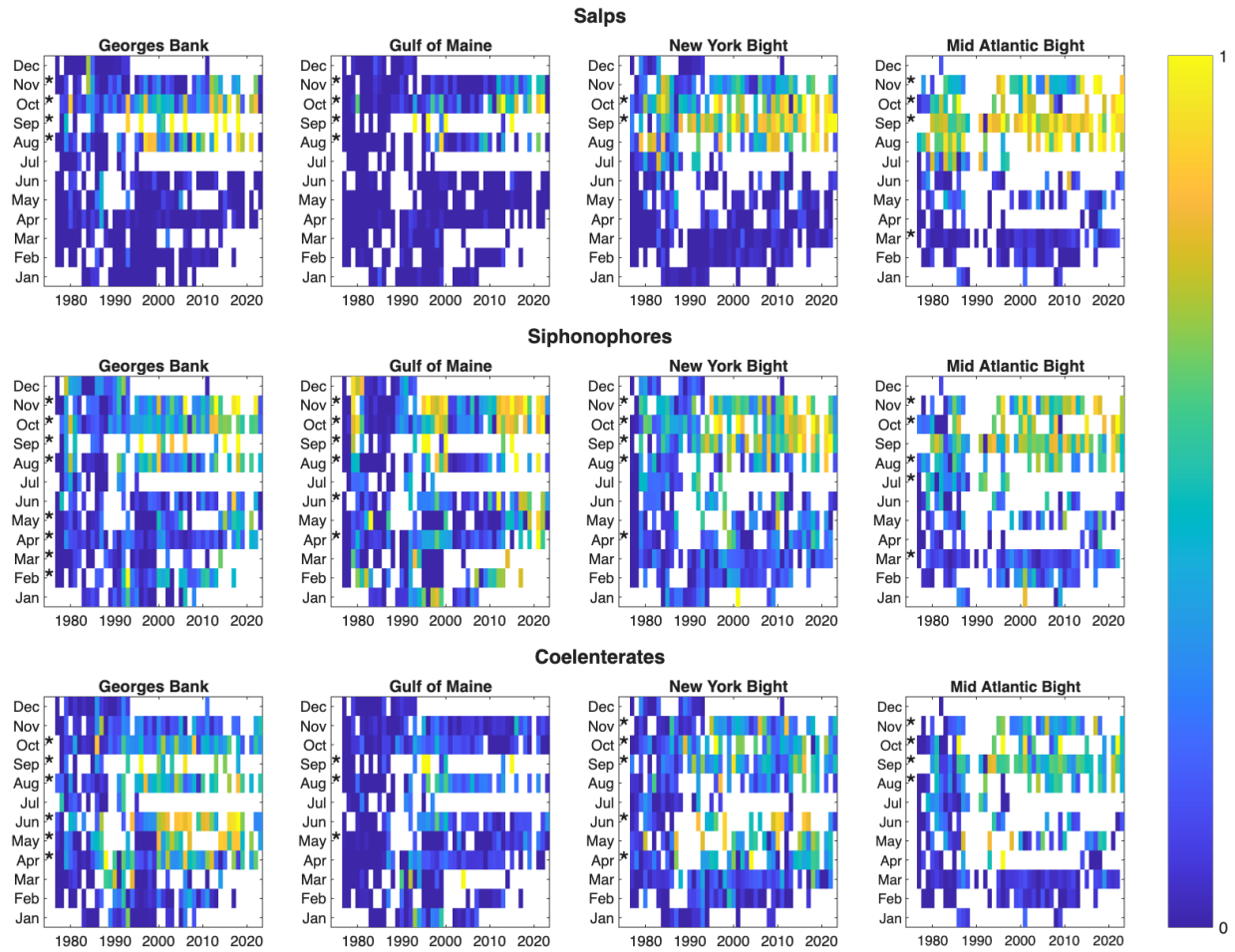


Figure 4: Figure S1. Time series of the proportion of presences of the indicated taxon broken down by month and regional polygon. Stars indicate a statistically significant ($p < 0.05$) increase over the duration of the time series.

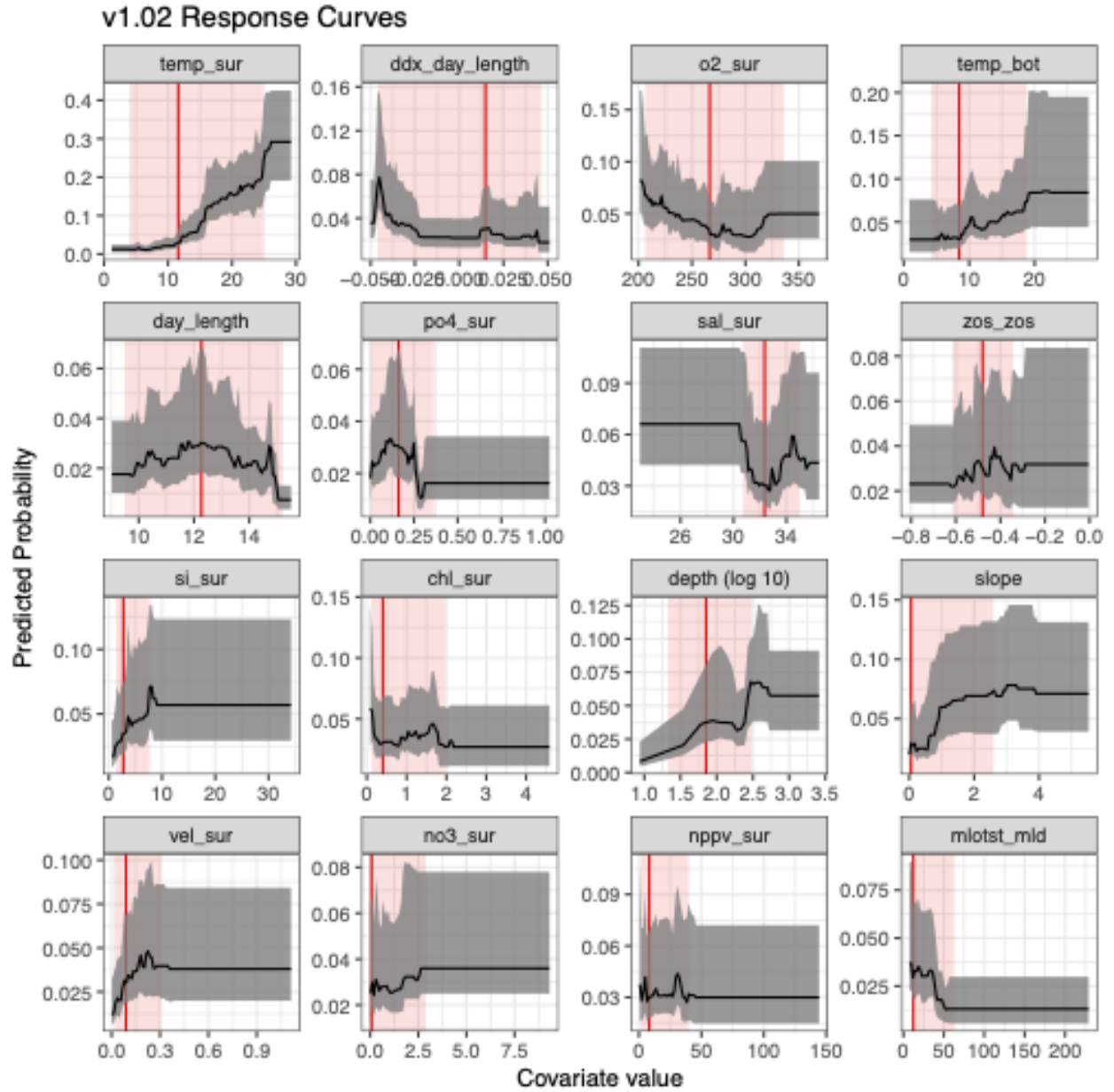


Figure 5: Figure S2. Response curves for salps. Red vertical lines indicate median value. Shaded area indicates 95% interquartile range of the response while other covariates are held at their median values. Response curves are ordered from left to right and top to bottom by variable importance.

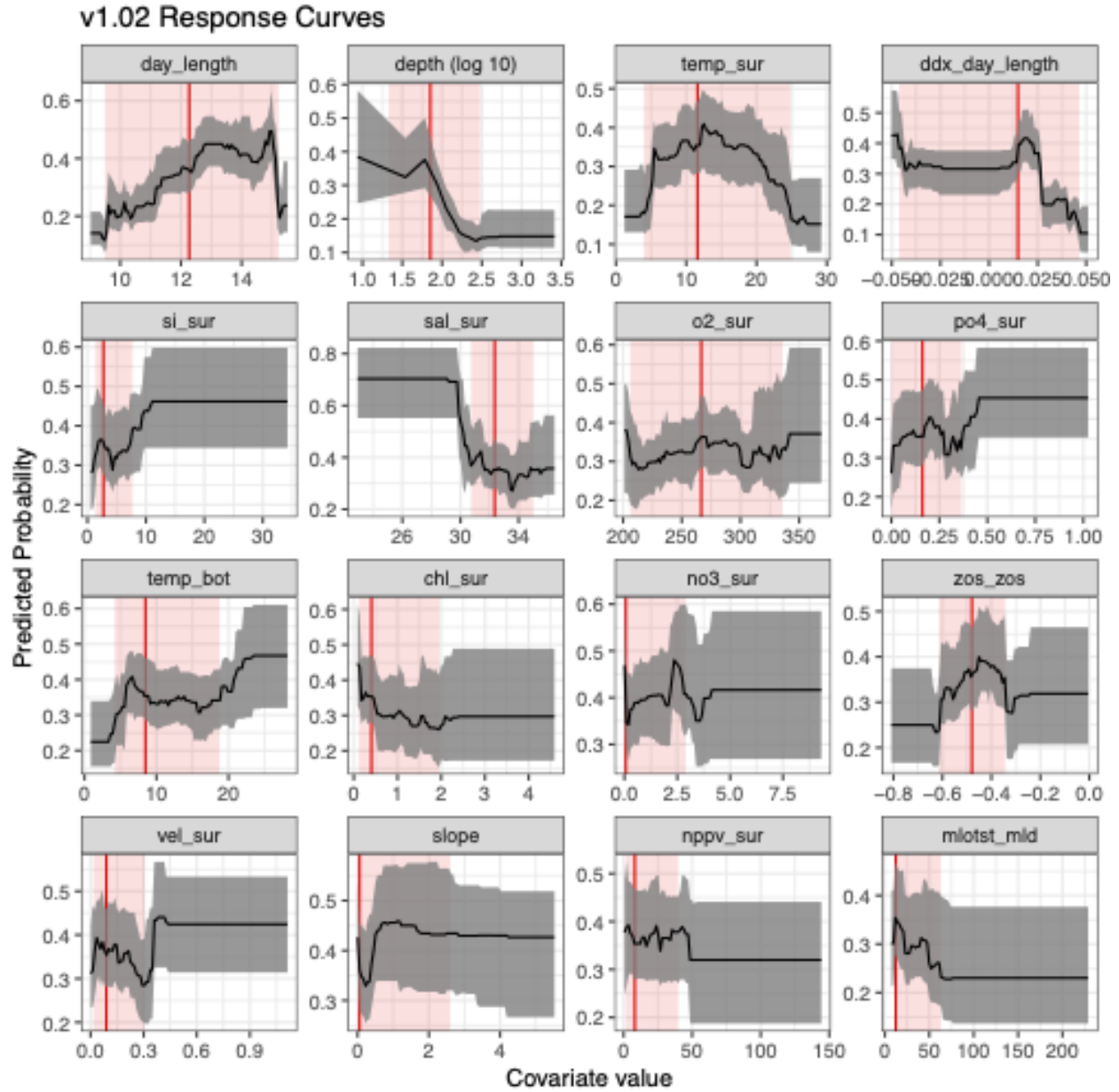


Figure 6: Figure S3. Response curves for coelenterates Red vertical lines indicate median value. Shaded area indicates 95% interquartile range of the response while other covariates are held at their median values. Response curves are ordered from left to right and top to bottom by variable importance.

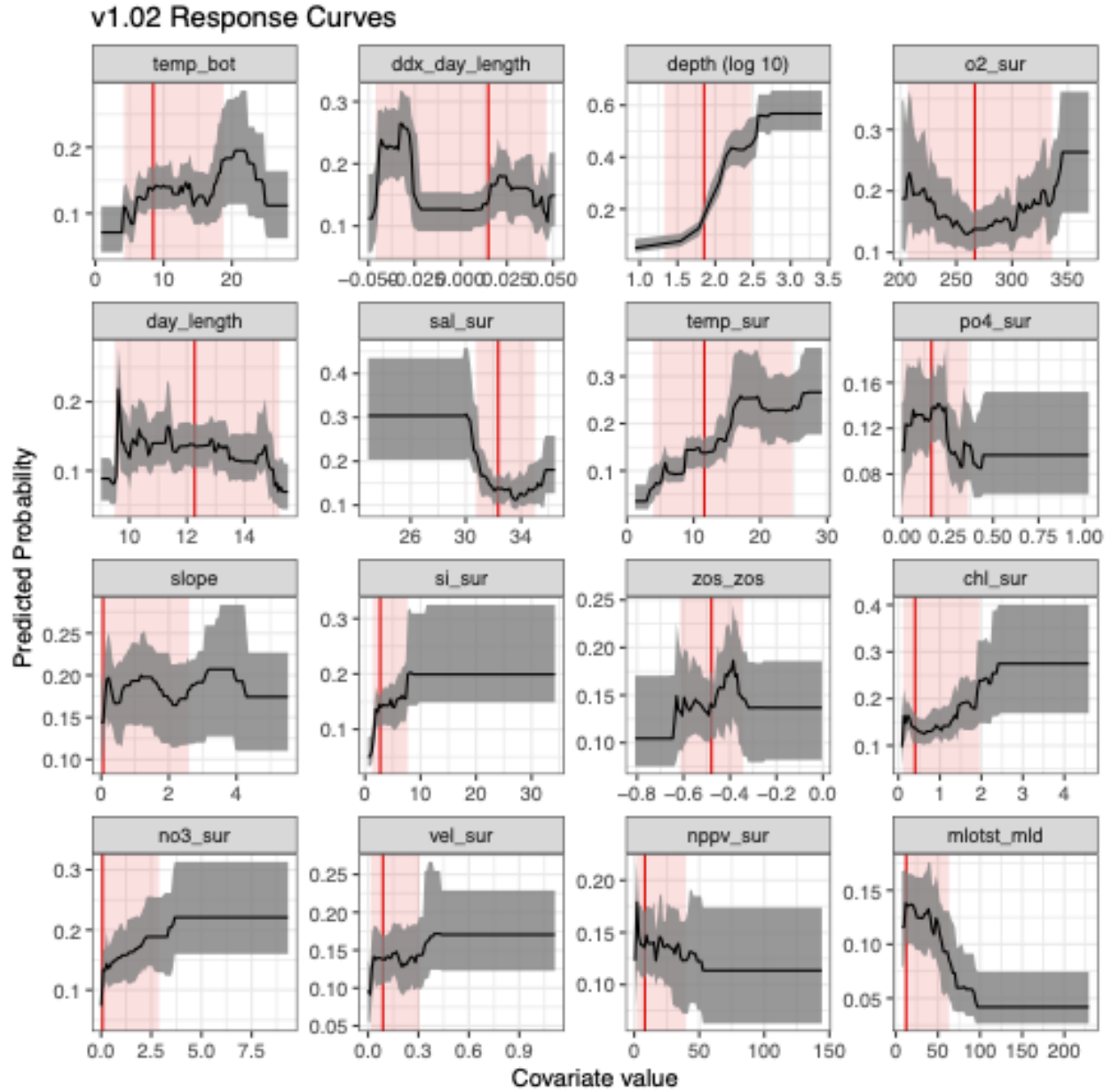


Figure 7: Figure S4. Response curves for siphonophores. Red vertical lines indicate median value. Shaded area indicates 95% interquartile range of the response while other covariates are held at their median values. Response curves are ordered from left to right and top to bottom by variable importance.

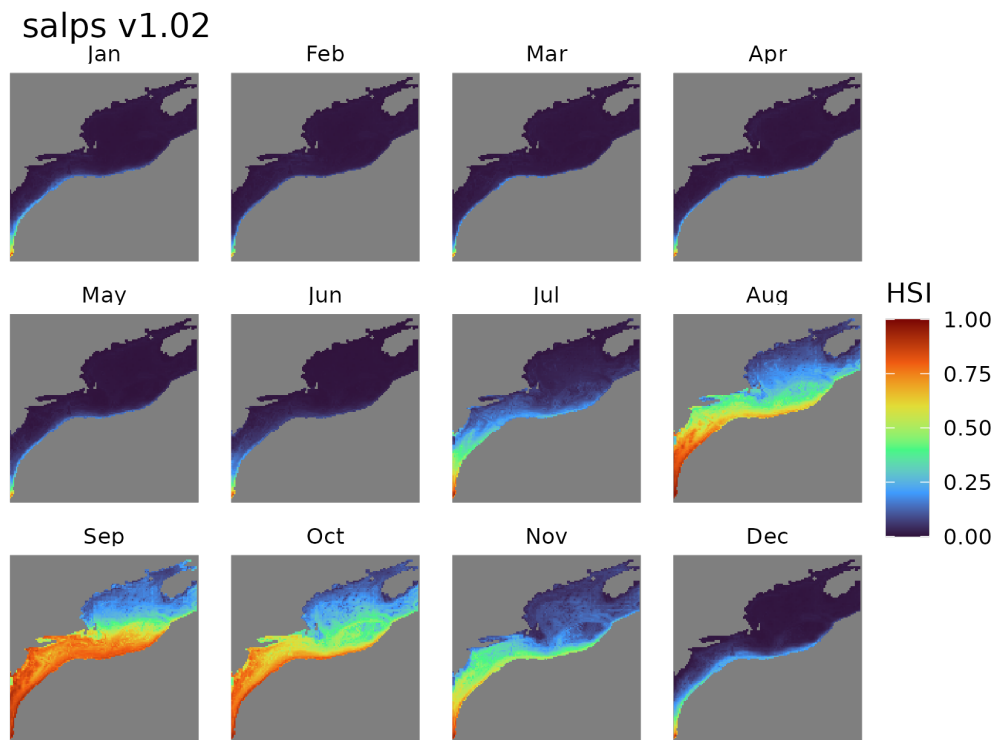


Figure 8: Figure S5. Model output climatology for salps.

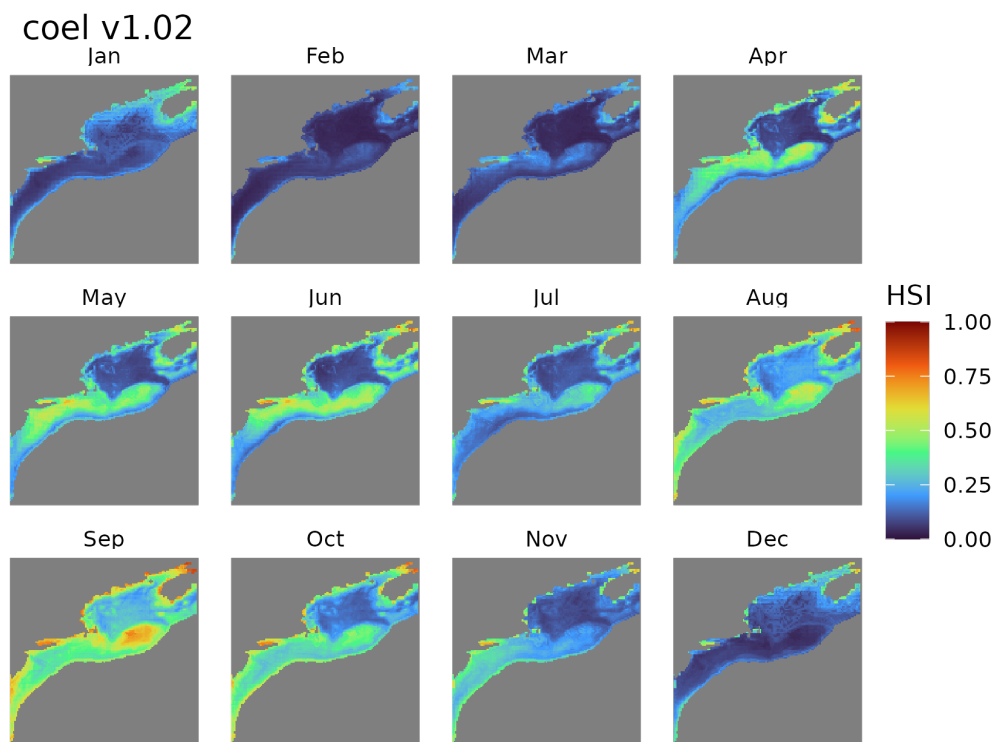


Figure 9: Figure S6. Model output climatology for coelenterates.

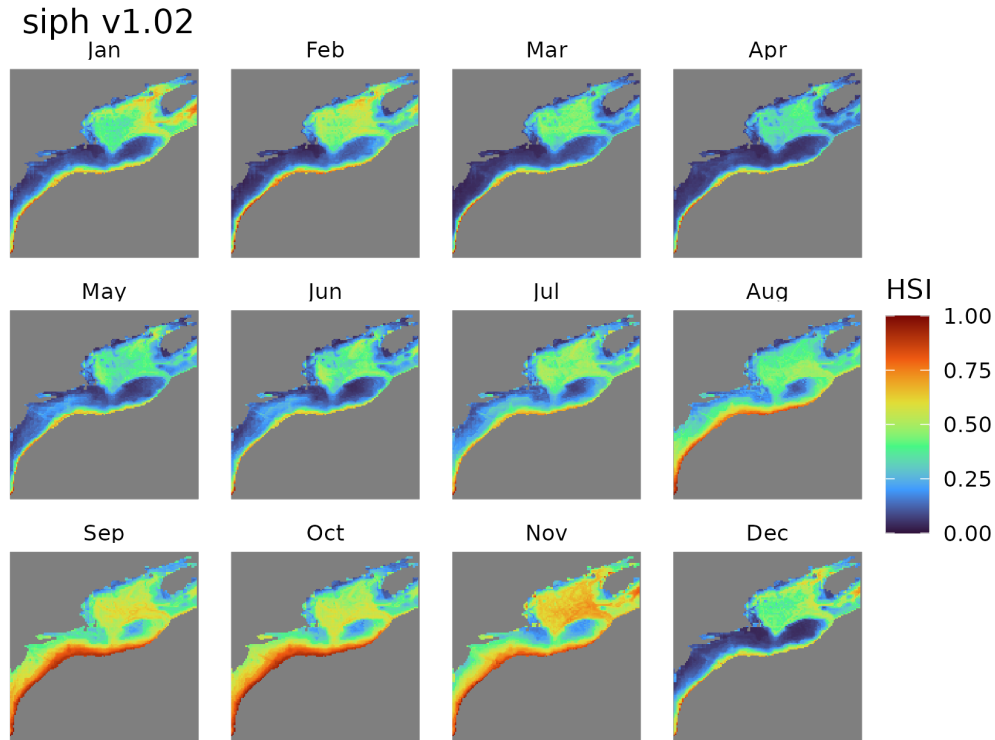


Figure 10: Figure S7. Model output climatology for siphonophores

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